

Question Paper

Exam Date & Time: 27-Mar-2025 (09:30 AM - 10:30 AM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Mid Term Examination - March 2025
CSE207: Design and Analysis of Algorithms

DESIGN AND ANALYSIS OF ALGORITHMS [CSE207]

Marks: 30

Duration: 60 mins.

Descriptive Only

Answer all the questions.

All Questions are Compulsory. Do As Directed.

- 1 ☐ ☐ ☐ Consider a sorting algorithm that places each element of an array in its correct position by comparing it with previously sorted elements and shifting them accordingly. Derive the worst-case recurrence relation for this algorithm and solve it using an appropriate method. Explain under what conditions this worst case occurs and analyze its impact on time complexity. (5)

Course Outcome: "CO1" / Cognitive Level: "Create"

- [OR] 2 ☐ ☐ ☐ Given two sequences, S1 and S2, where elements can be inserted or deleted to form a common pattern, determine the maximum-length sequence that appears in both S1 and S2 while maintaining relative order. Formulate a dynamic programming approach to compute the solution efficiently, derive its state transition relation, and analyze its time and space complexity. Given: S1 = 'ABCBADAB', S2 = 'BDCABA'. (5)

Course Outcome: "CO3" / Cognitive Level: "Create"

- 3 ☐ ☐ ☐ A company must assign four tasks to four available workers. Each worker has a different efficiency level for each task, represented by a cost matrix where the objective is to minimize the total assignment cost. Given the following cost matrix, use the Branch and Bound method to determine the optimal assignment of tasks to workers. Analyze the computational efficiency of the approach. (5)

Cost Matrix (Workers × Jobs):

	Job1	Job2	Job3	Job4
a	9	2	7	8
b	6	4	3	7
c	5	8	1	8
d	7	6	9	4

Course Outcome: "CO5" / Cognitive Level: "Application"

- [OR] 4 ☐ ☐ ☐ A treasure hunter explores an ancient cave and discovers a collection of valuable artifacts, each with a specific weight and worth. However, the hunter can only carry a limited weight due to the constraints of their backpack. Given that partial fractions of artifacts can be taken, determine the optimal selection strategy to maximize total value. What will be the worst case complexity for the approach? (5)

Given Items (Weight, Value):

Item	Weight	Value
1	10	60
2	20	100
3	30	120

Capacity: 50

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5 ☐ ☐ ☐ Solve the following recurrence with suitable method: (5)
- a. $T(n) = 2T(\sqrt{n}) + \log n$
- b. $T(n) = 3T(n-1) + n^4$

6 A high-performance computing system needs to efficiently multiply two large square matrices. Traditional matrix multiplication operates in $O(n^3)$ time, which becomes computationally expensive for large inputs. To optimize performance, an advanced divide-and-conquer approach is considered. Formulate an algorithm that recursively breaks down the problem into smaller subproblems and combines them using fewer multiplications than the standard method. Derive the recurrence relation for this approach, solve it using the Master Theorem, and analyze its time complexity. Finally, discuss the conditions under which this method is advantageous compared to the standard approach. (5)

7 The city of **Metroville** is upgrading its transportation system to ensure smooth traffic flow and reduce congestion. The city consists of multiple intersections (nodes) connected by roads (edges) with weights representing travel time and traffic density. Metroville's traffic department is considering various graph-based approaches to optimize connectivity and surveillance. **Problem Statement:** (10)

Metroville's road network is modeled as a **weighted, connected undirected graph**, where:

- **Intersections** (nodes) are labeled **P, Q, R, S, T, U**.
- **Roads** (edges) have **weights representing travel times (in minutes)**.

From	To	Travel Time (minutes)
P	Q	6
P	R	3
Q	R	2
Q	S	7
R	S	4
R	T	5
S	U	3
T	U	8

Attempt Any Two out of four and Write your Answer in brief. Each question carries equal marks

1. The city wants to construct a minimum-cost road network. Which algorithm, is more suitable for minimizing construction costs? Apply any one algorithms and show the spanning tree and total cost.
2. The mayor wants to determine the **shortest travel time** from **P to all other intersections**. Would **Dijkstra's or Bellman-Ford** algorithm be more suitable? What if some roads had **negative travel times** due to road closures or special traffic rules?
3. The city needs a system to **quickly determine the shortest travel time between all pairs of intersections** for emergency services. Compare the **Floyd-Warshall and Bellman-Ford** algorithms in terms of computational efficiency and suitability.
4. Metroville's traffic police need to **patrol all intersections efficiently** to ensure security and traffic rule compliance. Which graph traversal algorithm is better suited for different surveillance needs? Apply the suitable graph traversal for the surveillance team which needs to **visit every road exactly once** for maintenance checks.

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Event Name		Subject	
CHARUSAT Midterm Exam 2024-25 Even		CSE207:DESIGN AND ANALYSIS OF ALGORITHMS	
#	Course Outcome	Description	Marks
1	NA		10
2	NA		5
3	NA		5
4	NA		5
5	NA		5

5	NA		15
			Total
			40

* NA - CO Not Selected

Event Name		Subject	
CHARUSAT Midterm Exam 2024-25 Even		CSE207:DESIGN AND ANALYSIS OF ALGORITHMS	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(25%)
4	Analysis	1	5(12.5%)
5	Create	2	10(25%)
6	Evaluate	2	15(37.5%)
Total		7	40 (100%)